Model 2: XGBoost — methods and performance

Methods

- **Model:** XGBoost — many shallow trees fit in sequence; captures *non-linear* effects and *interactions*
- **Predictor sets:** 15 covariates vs. covariates + 12 behaviors
- **Split:** same 80/20 train/test (2,235/559), stratified on outcome
- **Missing data:** handled natively (no imputation)
- **Tuning:** randomized search, 60 settings × 5-fold CV → depth-3 trees, small learning rate, strong regularization
- **Added value (ΔR^2):** 2,000-sample test bootstrap + repeated 5-fold CV (25 folds)
- **Importance:** permutation (single and grouped), SHAP values and interactions

Performance (baseline test $R^2 = 0$)

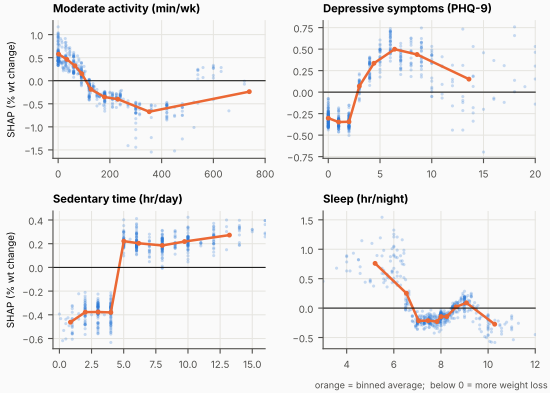
Predictors	Test R^2	CV R^2	Test RMSE
Covariates	0.065	0.127	8.783
+ **Behaviors**	**0.075**	**0.138**	**8.735**
ΔR^2 (test) [95% CI]	0.010	[−0.015, 0.033]	
ΔR^2 (CV mean)	0.011	(better in 22 of 25 folds)	

Grouped permutation importance (drop in test R^2)

BMI 1 yr ago alone	0.213
Other 14 covariates together	0.043
All 12 behaviors together	0.023

Behaviors add a small, fairly consistent gain; the test-set CI includes 0.

Model 2: what the model learned



SHAP dependence, top four behaviors by mean |SHAP| (test set).

Caution: PHQ-9 and sleep have large SHAP values but ≈0 test permutation importance, so their patterns may not generalize. All results are associations.

Key findings

- **Starting BMI dominates** (mean |SHAP| 2.55 vs. $\leq$ 0.47 for all others)
- **Behavioral effects are non-linear:**
  - Moderate activity: benefit mainly up to ∼150–300 min/wk
  - Sedentary time: threshold at ∼5 hr/day
  - Short sleep (<6.5 hr): less weight loss
  - PHQ-9 $\geq$ 3: less weight loss
- **Most reliable on new data:** moderate activity, then sedentary time, vigorous activity
- **Interactions:** strongest all involve starting BMI (BMI × moderate activity, BMI × PHQ-9)